AMAZON AI CYBER SECURITY HACKATHON · ROUND 1 · TRACK 2

Thehro

A pre-payment safety check inside a participating bank or UPI app that spots a coached scam payment before the PIN, and keeps interruptions to genuine payments inside a fixed budget.

> AI-Driven Scam Pattern Recognition: "Detect emerging AI-enabled scam workflows in UPI, banking, or wallet contexts and deliver timely, useful alerts without creating alert fatigue."

94%	0.89	148 / 157	0
of simulated scam victims warned	false pauses per 10,000 genuine payments	simulated mule campaigns flagged	genuine holds in the synthetic test

Team Name : [abhinavraj24**]**
Team Member : [Abhinav Raj] · [Abhinav Kumar] · [Megha Kumari] · [Iqra Afreen] ·
Institute: [IIT KANPUR]

Synthetic data only · not yet validated on real payments

A scam victim pays with their own PIN, so every payment looks normal

PROBLEM AT A GLANCE

Who we protect

UPI and bank-app customers being talked through a multi-step scam. We start with people aged 55+, who lost ₹390,000 per complaint in 2025.

The threat: AI-enabled scam workflows

AI makes the script cheap and convincing: cloned voices of relatives, fake "police" on video calls, chatbot-run trading and task groups. The money still moves as rising payments to new payees.

The harmful moment

The seconds before the next payment. Mule accounts move the money on within minutes, so it is rarely recovered.

THE NUMBERS

~7

frauds reported per million UPI payments, FY 2024-25

Lok Sabha Q2481, Dec 2025

76%

of 2025 cyber-fraud losses came from investment scams

NCRP 2025 data, via ThePrint

Why we watch the payment: Thehro is not a deepfake detector. AI can fake a voice, a video or a chat, but every scam ends in a payment, so Thehro looks for the risky payment pattern instead.

So the real design limit is alert fatigue: a warning everyone sees is a warning everyone ignores.

Senior-loss figure: Rajya Sabha reply, via Asianet. All sources and assumptions: Appendix C and E.

Today a scam shows up in a complaint. Thehro acts before the PIN.

TYPICAL GAP each payment judged alone	**Hook** AI voice, video call or chat group	**Coached payment** Right PIN, own phone	**Mule account** Money moved on in minutes	**Generic warning** Shown to all, so ignored	**Complaint** Days later, if ever
WITH THEHRO whole journey judged	**Same hook** No chat or call content read	**Check before PIN** Journey, payee and call context	**Pause with facts** 3 reasons and a way out	**Escalate** Trusted person or bank officer	**Record** Evidence log, 1930 draft

CONVENTIONAL	THEHRO
Payments are often scored one at a time, so a coached payment can look normal.	Scores the journey: payer history, payee risk and live call context together.
Some mule networks are found only after victims report, days later.	A label-free detector links mule accounts from money flows, within hours in simulation.
Warnings go to everyone, so people tap past them.	Pauses only when two kinds of evidence agree, inside a 1-in-10,000 budget.

Sources: TechCrunch (Google Pay warnings) · LocalCircles (unreported fraud) · Timings from our prototype

Five steps before the PIN; one signal can warn but never hold money

STEP 1

Signals

Journey from bank records, payee risk, opt-in phone flags

39 features

STEP 2

Score

A model per family plus one fused score, trained on simulated reported cases

~25 ms

STEP 3

Link mules

Accounts that collect from new payers and pass money on fast

no labels

STEP 4

Gate

Delaying money needs 2 of 3 families above their usual range

2 must agree

STEP 5

Tier

Lines set so genuine payments cross each at a fixed rate

budgeted

THE FIVE TIERS · ONE TABLE EVERYWHERE

T0 · Allow

Below every line. Nothing shown.

T1 · Note

One line under Pay; money moves. Budget: 1 in 1,000.

T2 · Pause

Facts and advice; the customer may Continue after 30 s. Budget: 1 in 10,000.

T3 · Trusted check

2 families + ₹50,000+. Waits for a trusted person, at most 1 hour.

T4 · Hold

2 families + payee flag + ₹10,000+. Not sent until an officer decides.

After a customer continues past a warning: for 7 days a note becomes a pause and the ₹ lines are waived.

"Above usual range" = over the 99th percentile of genuine payments. Every tier acts before the PIN: no money has left yet.

05 / ONE JOURNEY, END TO END

₹15,693 to a new account, 70 minutes into a call

Synthetic test-day journey

1 · INPUT

Journey
First payment to this account, 55× her usual amount

Payee
Opened 58 days ago; in a live network of 9 accounts; passed on 95% of ₹234,000 in a day

Context (opted in)
Internet call from an unknown number, 70 minutes so far

2 · WHAT SHE SEES

We're holding this payment to protect you

Is someone on a call telling you to pay?

Don't pay because a caller says they are police. Hang up; call your bank or 1930.

Cancel payment

Talk to my bank

3 · DECISION

Fused risk score

0.38



Pause line: 0.085

3 of 3 families agree

Payee flag: live mule network

Tier 4 · Hold for review

The payment is not sent, so no money leaves her account. An officer releases or cancels it within 30 minutes.

4 · EVIDENCE RECORD

```
record #0 · tier 4
reasons:
CAMPAIGN
NEW_PAYEE_LARGE
CALL_UNKNOWN
hash da4705303787…
prev 000000000000…
```

The officer's outcome and a 1930 complaint draft for her to file are appended. Editing any record breaks every later hash.

Digital-arrest journey from our synthetic test days, customer aged 60+. Her next two requests reached tiers 4 and 3.

Simulation: 94% of victims warned at 0.9 false pauses per 10,000

Synthetic results

94.3%
of simulated victims warned
95% CI 92.4–95.8 · n = 720

0.89
false pauses per 10,000
95% CI 0.69–1.12 · 786,121 payments

69.6%
warned at the first payment
95% CI 66.1–72.8

0
genuine holds observed
95% bound 1 in 213,000; live rate unknown

AGAINST SIMPLE RULES, SAME TEST DAYS

Approach	Pauses /10k	Warned
Thehro	0.89	94%
Pause every first ₹10k+ to a new payee	59.8	69%
Pause when the payee is on a fraud list	6.1	43%
Pause during calls from unknown numbers	5.1	28%

Estimated, not measured: at India's assumed fraud rate, about **1 in 9** Thehro pauses would be a real scam (scam pauses ÷ all pauses); for the first rule, 1 in 1,076.

MODELLED, NOT MEASURED

Money protected: 93% if warned victims stop at assumed rates, 76% if half as many stop. A warning is not a prevented loss.

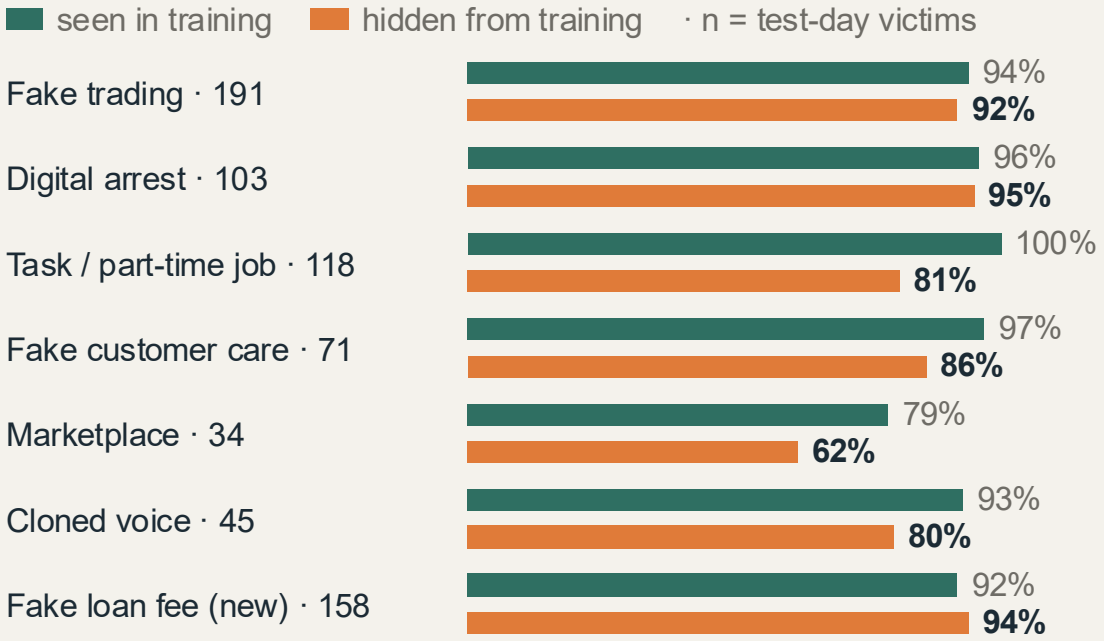
HOW WE TESTED

40,000 synthetic customers, 60 days, 2.78M payments. Trained on days 14–34, lines set on 35–41, tested on 42–59. The pause interval crosses 1 per 10,000: under budget on average, not guaranteed.

Warned = paused (T2) or higher. Split by time only, so customers and payees recur. Baselines are fixed, untuned rules.

New scam types are still caught: the warning signs are shared

VICTIMS WARNED WHEN A TYPE IS HIDDEN FROM TRAINING



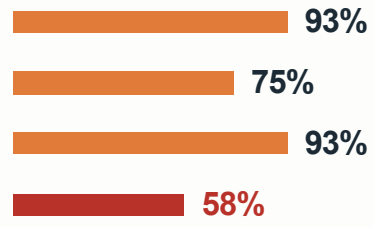


148 of 157

synthetic scam campaigns flagged by the graph heuristic

Median 16 h after the first victim paid; 78% before the first complaint

If scammers adapt (same model, not retrained; 720 victims)



92%

of fake loan-fee victims warned with 2 reported cases in training, and 94% with none (n = 158). Customers and payees can recur across the time split.

All seven types come from our simulator: this tests shared warning signs, not unseen real attacks.

Consented, explainable, appealable. A held payment waits for a person.

Data and consent

- Uses the bank's payment history, payee risk scores and yes/no flags computed on the phone (opt-in)
- Never collected: call audio, messages, contacts, location
- Consent can be withdrawn in settings; it stops at the next payment

Missing data and failures

- A missing field or failed payee feed counts as no flag, never as risk
- The decision never waits for a slow feed
- If the service is down, the bank's current rules apply and nothing is held

Overrides and appeals

- Continue past a warning: 7 days of closer checks for that customer
- A payee still clean 7 days later stops triggering warnings
- "This was me" opens a case; an officer can release a hold and confirm the payee

Human review and misuse

- A held payment is never sent until an officer releases it (30-minute target, 24×7)
- No answer: the payment is cancelled and the money stays in the account
- Lines never published; evidence log hash-chained and access-logged

Fairness check (synthetic): age, language and region are never inputs; 91–97% of victims warned across age bands.

Context flags are kept only in tier 2+ evidence records; their retention period is set with the bank's legal team.

Owned by the payer's bank: start silent, widen only past each gate

PHASE 1 · 4 WEEKS

Shadow mode

Score live payments, show nothing

Go/no-go: ≤1 false pause per 10,000, overall and in every age, language and region group, on adequate samples

Not yet met in our run: ages 41–59, no opt-in, Gujarati (A1)

→

PHASE 2

Opted-in, 55+

Notes and pauses only, A/B test; 55+ is our pilot choice

Gate: bank security and compliance sign-off; stop rate measured; kill switch and named rollback owner

→

PHASE 3

All customers

Add trusted checks and holds; staff the officer queue

Gate: every hold reviewed within 30 minutes

→

PHASE 4

Every bank

Share mule-network flags through NPCI and IDPIC

Needs: data-sharing agreements

Owner and dependencies

The bank's app calls our API before the PIN. Needs a participating bank app, payee-feed agreements and a security review. Phone flags only via an approved integration; the model runs without them. Without the feeds it runs on the bank's own data and network flag.

Regulatory context (our reading)

RBI's June 2026 directions (from Jan 2027) raise banks' accountability for electronic-banking fraud; specific, logged warnings help show what a bank did. Our ₹10,000 and ₹50,000 lines echo RBI's April 2026 discussion paper; they are not RBI rules.

Per 10 million payments a day

About 890 genuine and 117 scam payments paused (1 in 9 real, as on slide 6), including about 65 that reach a T4 hold. Scenario estimates, not live results.

Coverage: participating apps only; third-party UPI apps would need their own integration or a shared risk service (not claimed today).

Finale: three workstreams, each with a pass line

Detection — 0–20 h

BUILT

Simulator, 4 models, network detector, policy engine

TO BUILD

- Pin versions; rebuild from scratch
- Harder world: split payments, aged mules, chat-only coaching

PASS IF

≥85% of victims warned at ≤1 false pause per 10,000, with 95% CIs

Customer and reviewer — 8–32 h

BUILT

EN/HI messages, REST API, evidence log

TO BUILD

- Android SDK stub: consent screen, "Is someone on a call?" prompt, mock PIN screen
- Officer console with the hold queue
- Failure paths: feed timeout, no consent, no officer

PASS IF

p99 under 100 ms over 4G; consent off sends no flags; every failure falls back to the lighter action

Red team — 32–48 h

BUILT

20 automated tests, 4 stress tests

TO BUILD

- Line probing, amount splitting, mule rotation
- Switch off network and phone signals in turn
- Fairness re-check; freeze and rehearse

PASS IF

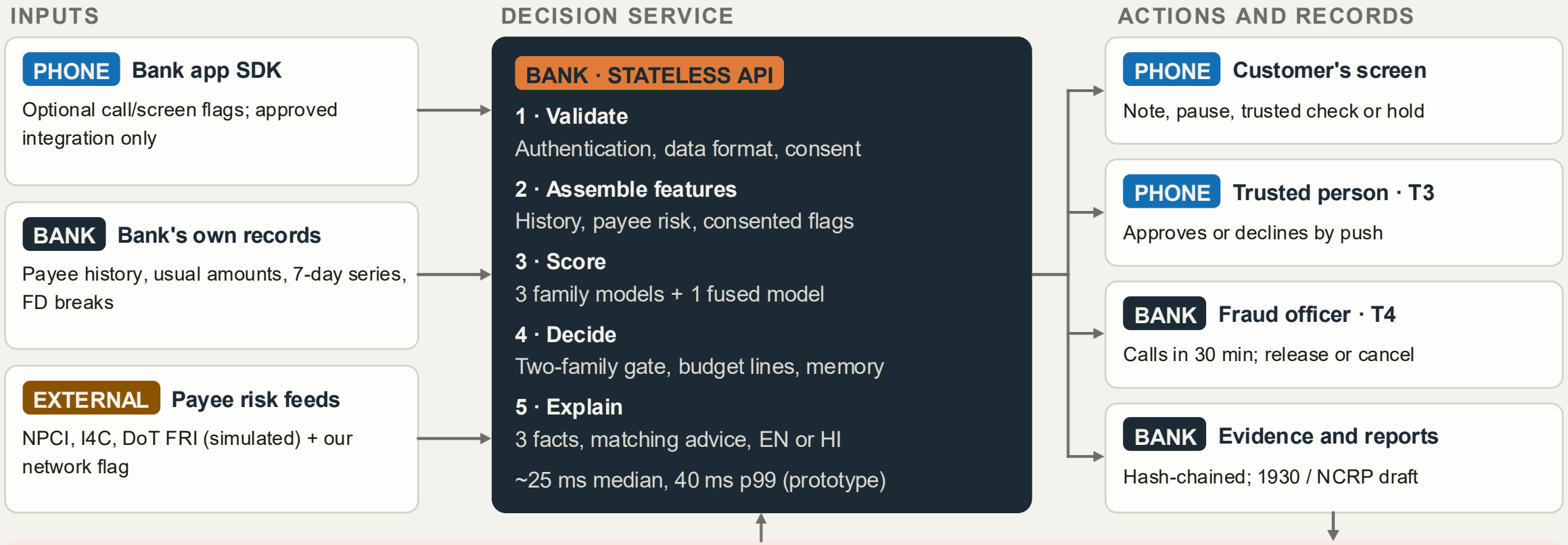
No tactic drops warned below 75% of the 720 test victims; each signal's gain reported; no age group over 1.5× the pause rate

Assumptions we test: feeds answer in time · 70% opt in · 35–75% of warned victims stop · synthetic-to-live transfer unverified

Demo scope: mock pre-payment flow, synthetic inputs, risk decision, warning and evidence record in under 5 minutes. Live feeds, other apps and 24×7 staffing come later.

ARCHITECTURE 1 OF 3 / END-TO-END SYSTEM

One decision before the PIN: three inputs in, four actions out



Learning loop Officer outcomes and complaints become labels, days late → weekly retraining → lines re-set to the budgets → versioned release.

PHONE = on the customer's phone · BANK = bank-side · EXTERNAL = outside feeds, simulated by our data generator

What goes in, what is kept, and who acts at each tier

Input	Source and provenance	Consent	Kept
Journey history	The bank's own ledger and deposit records	Bank's fraud-monitoring duty	Bank records, under its existing rules
Payee risk	NPCI, MuleHunter.AI, I4C, DoT FRI (simulated now); our network flag	Data-sharing agreements; each feed optional	Scores only; a failed feed = no flag
Context flags	Approved integration only; else absent	Opt-in; withdraw any time; off = no vote	Only in tier 2+ records, else dropped
Outcomes and evidence	Customer choice, officer decision, complaint	Role-based access, reads logged	Pseudonymous, hash-chained

Components

- Feature store: rolling windows per customer and payee
- Models: 3 family + 1 fused gradient-boosted trees
- Mule-network detector: streaming, label-free graph
- Policy engine: two-family gate, budget lines, memory
- Evidence log: hash-chained (in-memory in prototype)
- Interfaces (bank backend, mTLS): /v1/score, /v1/outcome, /v1/evidence

Escalation, review and reporting

- T2: Cancel ends it; Call my bank dials the saved number; Continue starts the 7-day memory
- T3: a nominated trusted person approves by push; else a 1-hour wait, then the customer decides
- T4: request not sent to UPI; customer told in app and by SMS; officer calls in 30 min; no answer = cancelled
- Officer: release and confirm the payee, or cancel, pre-fill 1930 / NCRP, report the mule

Every tier 2+ decision and outcome is appended to the evidence log. Its retention is set with the bank's legal team.

Threats met in layers; every number stated with its limits

Threat	Mitigation
Victim coached to ignore warnings	Checkable facts, then a trusted person or an officer; memory escalates after Continue
Payments split under a line	7-day series features; lines never published; ₹ lines waived after a warning
New mule per payment, or chat-only coaching	Network detector, account age, fan-in; journey and payee still decide (75% and 93% warned*)
Forged calls or spoofed flags	mTLS and signed requests from the bank backend; device attestation; context only raises risk
Insider misuse of evidence	Pseudonymous ids, role-based officer access, every read logged; the hash chain exposes edits
Feed or service outage	Never waits: missing feed = no flag; service down = bank's current rules, nothing held
Drift as scams change	Weekly retraining; lines re-set to budgets; rates watched by group; roll back if over budget

Performance

25 ms median, 40 ms p99 over 1,000 decisions, one Python process on 2 vCPU: a prototype figure, not an SLA. Target under 100 ms p99. Next: load-test until p99 crosses 100 ms.

Validation

Lines set on about 300,000 genuine payments, then fixed. 95% CIs on every rate. One week is too small for a 1-in-100,000 hold line, so the gates carry holds: 0 genuine holds observed (synthetic).

* Synthetic stress tests, 720 victims. Built: gates, network detector, hash chain. Proposed: mTLS, attestation, access roles.

Results by scam type and by customer group

Synthetic test days. Labels: training cases (by day 38). Hidden: type removed from training. Protected: modelled.

Scam type	Journeys	Labels	Median loss	Warned	At 1st	Protected	Hidden
Fake trading / IPO	191	176	₹129,567	94%	57%	96%	92%
Digital arrest	103	139	₹166,940	96%	91%	88%	95%
Task / part-time job	118	179	₹29,280	100%	90%	98%	81%
Fake customer care	71	86	₹66,288	97%	85%	89%	86%
Marketplace "scan to receive"	34	61	₹14,169	79%	29%	87%	62%
Cloned voice of a relative	45	34	₹51,123	93%	91%	82%	80%
Fake loan fee (new type)	158	2	₹19,494	92%	52%	84%	94%

Group	18–25	26–40	41–59	60+	Metro	Tier 2	Rural	Opt-in	No opt-in
Victims warned	95%	91%	97%	97%	96%	91%	97%	94%	94%
False pauses /10k	0.85	0.78	1.10	0.85	0.83	0.93	0.91	0.82	1.06

Language (victims)	Hindi (319)	English (140)	Tamil (68)	Telugu (10)	Bengali (71)	Marathi (73)	Kannada (33)	Gujarati (6)
Victims warned	93%	98%	91%	90%	97%	93%	91%	100%
False pauses /10k	0.99	0.91	0.95	0.76	0.77	0.72	0.26	1.32

Small groups have wide intervals: marketplace (n = 34) 79% warned, 95% CI 63–90%. Protected is modelled.

APPENDIX B / BUDGET TRADE-OFF AND NETWORK DETECTOR

More interruptions buy little; mule networks are found within a day

Pause budget = share of genuine payments allowed to reach tier 2+, set on about 300,000 validation-week payments

Budget	Pauses /10k	Warned	Money (model)	Scams per warning
1 in 100,000	0.28	83%	82%	1 in 4
3 in 100,000	0.45	91%	90%	1 in 5
1 in 10,000 (ours)	0.89	94%	93%	1 in 9
3 in 10,000	2.28	97%	95%	1 in 20
1 in 1,000	7.79	98%	96%	1 in 62
3 in 1,000	27.0	99%	96%	1 in 205
1 in 100	97.3	100%	96%	1 in 722

Label-free mule-network detector, per scam campaign (157 in all)

Scam type	Runs	Found	Median hours	Before complaint
Fake trading	26	96%	23	96%
Digital arrest	30	93%	20	73%
Task / job	22	100%	6	100%
Customer care	20	90%	7	55%
Marketplace	20	95%	26	80%
Cloned voice	25	92%	26	56%
Fake loan fee	14	93%	11	86%

Runs = scam campaigns. Median hours from a campaign's first victim payment to its network being flagged.

Every number we used: fact, estimate, assumption or measurement

Item	Kind	Value used	Source (Appendix E)
Reported UPI fraud rate	Fact	6.8 reported incidents per million payments: 1.26 million incidents in 185.9 billion payments, FY 2024-25	[1] Lok Sabha Q2481; [2] IANS
True fraud rate	Scenario	Reported × ~2 (51% never complain → ×2.04; we use ×2) ≈ 13.6 per million; one incident = one payment	[3] LocalCircles survey
Victims who stop when warned	Assumption	35–75% at a pause, 75–90% at a trusted check, 92–97% at a hold	Anchored on [4] CeDEx 2023
Opt-in and drop-off	Assumption	70% opt in; 8%, 3%, 1% of genuine payers give up at tiers 2, 3, 4	Ours; to be measured in phase 2
RBI proposals	Fact (proposal)	Proposed: one-hour lag at ₹10,000+; trusted person above ₹50,000 for 70+ and persons with disabilities; hold only the disputed amount. Our lines and 55+ pilot are our choices	[5] RBI paper, Apr 2026; [6] draft mule procedure, Sep 2026
Scam mix and targets	Fact	Investment scams 76% of losses; seniors lose ₹390,000 per complaint	[9] ThePrint; [10] Asianet
Time limit	Fact	UPI request-pay limit of 15 seconds	[8] MediaNama
"1 in 9" and slide 9 load	Scenario calc	Per million: 13.6 scam × 86% paused = 11.7; genuine 89; 11.7 ÷ 100.7 ≈ 1 in 9. Slide 9 = × 10; holds: 136 × 48% at T4 ≈ 65	Our pipeline output and [3]
Prototype results	Measured (synthetic)	Every result on slides 1, 5–7 and A1–A2	Our pipeline's saved output, one run

86% = scam payments paused (payment level); 94.3% = victims warned at least once (victim level). [n] = Appendix E.

APPENDIX D / DISCLOSURES

What we used, what we built, and how AI was used

Data

Fully synthetic: 40,000 customers, 60 days, 2.78 million payments and 1,675 scam journeys from our own simulator, calibrated to the public figures in Appendix C. No real customer, bank or platform data.

Models

Gradient-boosted trees from scikit-learn, trained from scratch on the synthetic data, and a label-free graph detector we wrote. No pre-trained or third-party models.

Third-party assets

Open-source libraries: numpy, pandas, scikit-learn, FastAPI, Pydantic, uvicorn, matplotlib, pytest. Carlito font (SIL Open Font License) and Arial. No logos or names of real banks or apps.

AI-generated content

Code, demo, deck text and research were drafted with Claude (Anthropic) and reviewed by the team; a ChatGPT (OpenAI) review informed revisions. Figures are taken from our pipeline's saved output or a cited source.

Code and tests: [repository link]. The deck stands alone without it.

APPENDIX E / REFERENCES

Where each external fact comes from

[1] Lok Sabha, unstarred question 2481: UPI fraud cases and volumes, FY 2024-25 (Dec 2025) · eparlib.sansad.in

[2] IANS: UPI transaction growth and volumes (Aug 2026) · ianslive.in

[3] Business Standard: LocalCircles survey on UPI fraud and non-reporting (Jun 2025) · business-standard.com

[4] CeDEx discussion paper 2023-08, University of Nottingham: friction and scam payments · nottingham.ac.uk

[5] Reserve Bank of India: discussion paper on digital payment fraud controls (Apr 2026) · rbi.org.in

[6] Taxguru: RBI draft procedure for mule accounts (Sep 2026) · taxguru.in

[7] ELP: summary of RBI customer-liability directions for electronic banking fraud (Jun 2026) · elplaw.in

[8] MediaNama: NPCI UPI changes, 15-second request-pay limit (Jun 2025) · medianama.com

[9] ThePrint: cybercrime losses in 2025, mainly investment scams (2026) · theprint.in

[10] Asianet Newsable: senior citizens' cyber-fraud losses in 2025, citing a Rajya Sabha reply · asianetnews.com

[11] TechCrunch: Google Pay scam warnings in India (Nov 2025) · techcrunch.com

[12] Department of Telecommunications: Financial Fraud Risk Indicator (FRI) · sancharsaathi.gov.in

[13] Taxguru: Parliament reply on MuleHunter.AI and the Digital Payment Intelligence Platform · taxguru.in

Links open the source; every number is also stated in the deck, so it reads without them.